PACKET SNIFFING USING RAW SOCKETS IN PYTHON

AIM:

To implement a packet sniffer using raw sockets in python to elaborate capture live network traffic and parse ethernet, IP and transport layer (TCP/UDP/ICMP) headers.

PROCEDURE:

1) Install python 3 on a linux system or virtual machine.
2) Open a terminal with root (sudo) privileges.
3) Create a python file named sniffer.py.
4) Import the required modules (socket and struct).
5) Create a raw socket using AF_Packet and sock_RAW.
6) Capture incoming ethernet frames from the network.
7) Parse the ethernet header to obtain source MAC, destination MAC and protocol type.
8) If the protocol is IPv4, parse the IP header.
9) Extract IP version, header length, TTL, protocol number, source IP, and destination IP.
10) Display the captured packet information on the screen.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sanja>python --version
Python 3.10.0

C:\Users\sanja>mkdir PacketSniffer

C:\Users\sanja>cd PacketSniffer

C:\Users\sanja\PacketSniffer>cd PacketSniffer
The system cannot find the path specified.

C:\Users\sanja\PacketSniffer>type nul > sniffer.py

C:\Users\sanja\PacketSniffer>dir
 Volume in drive C is Windows
 Volume Serial Number is 2A91-9DD6

 Directory of C:\Users\sanja\PacketSniffer

31-07-2026  18:56    <DIR>          .
31-07-2026  18:55    <DIR>          ..
31-07-2026  18:56                 0 sniffer.py
               1 File(s)              0 bytes
               2 Dir(s)   3,026,112,512 bytes free
```

```
Select Administrator: Command Prompt - python sniffer.py
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\Users\sanja\PacketSniffer

C:\Users\sanja\PacketSniffer>python sniffer.py
Packet Sniffer Started...

--------------------------------------------
Version : 4
Header Length : 20
TTL : 128
Protocol : 6
Source IP : 10.51.77.107
Destination IP : 10.51.77.44
Packet Length: 52
--------------------------------------------
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Header Length : 20
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Packet Length: 52
--------------------------------------------
```

File Edit View

```
import socket
import struct

# Create raw socket
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

# Local machine IP
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))

# Include IP headers
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)

# Enable promiscuous mode (Windows)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

print("Packet Sniffer Started...\n")

try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)

        ip_header = raw_data[0:20]
        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)

        version = iph[0] >> 4
        ihl = iph[0] & 0xF
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])

        print("----------------------------------------")
        print("Version :", version)
        print("Header Length:", ihl * 4)
        print("TTL :", ttl)
        print("Protocol :", protocol)
        print("Source IP :", source)
        print("Destination IP :", destination)
        print("Packet Length:", len(raw_data))

except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

Ln 44 Col 23 1,223 characters Plain text

Result:-

The packet sniffer was successfully implemented using python raw sockets. It captured live network packets and correctly extracted & displayed the ethernet and IPv4 header information, including MAC address, IP addresses, TTL, and Transport layer protocol details.